Square Roots

Simple Notes, Examples, Methods & Q\&A (100 - 200)

A square root is the reverse of squaring a number. This guide explains the idea in simple words with examples, the main methods to find square roots, and ends with a set of solved questions for numbers between 100 and 200.

1. What is a Square Root?

Definition: The square root of a number is the value which, when multiplied by itself, gives that number. It is the opposite (inverse) of squaring.
Symbol: The radical sign sqrt( ) is used. Example: sqrt(25).
Examples:
- sqrt(25) = 5 because 5 x 5 = 25
- sqrt(81) = 9 because 9 x 9 = 81
- Squaring: 6 x 6 = 36 -> Square root: sqrt(36) = 6

2. Perfect Squares

Definition: A perfect square is a number whose square root is a whole number.
Examples: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 ...
Notes:
- Square roots of perfect squares are exact whole numbers.
- Numbers like 2, 3, 5, 7 are NOT perfect squares; their roots are non-ending decimals (irrational), e.g. sqrt(2) = 1.414...
- A perfect square never ends in 2, 3, 7 or 8.

3. Methods to Find a Square Root

(a) Prime Factorisation: Break the number into prime factors, make pairs, and take one factor from each pair.
- Example: 144 = 2x2 x 2x2 x 3x3 -> take 2 x 2 x 3 = 12, so sqrt(144) = 12

(b) Long Division: Used for large numbers and non-perfect squares to get the answer step by step (even in decimals).
(c) Estimation: Find the two nearest perfect squares to guess the approximate value.
- Example: sqrt(150) lies between sqrt(144)=12 and sqrt(169)=13, so it is about 12.2

4. Useful Properties

- sqrt(a x b) = sqrt(a) x sqrt(b)
- sqrt(a / b) = sqrt(a) / sqrt(b)
- sqrt(0) = 0 and sqrt(1) = 1
- A negative number has no real square root.

5. Perfect Squares Between 100 and 200

There are only five perfect squares in this range:

sqrt(100) = 10 (10 x 10 = 100)
sqrt(121) = 11 (11 x 11 = 121)
sqrt(144) = 12 (12 x 12 = 144)
sqrt(169) = 13 (13 x 13 = 169)
sqrt(196) = 14 (14 x 14 = 196)

6. Questions & Answers (100 - 200)

Q1. What is sqrt(100)?
- Ans: 10 (because 10 x 10 = 100)

Q2. What is sqrt(121)?
- Ans: 11 (because 11 x 11 = 121)

Q3. What is sqrt(144)?
- Ans: 12 (because 12 x 12 = 144)

Q4. What is sqrt(169)?
- Ans: 13 (because 13 x 13 = 169)

Q5. What is sqrt(196)?
- Ans: 14 (because 14 x 14 = 196)

Q6. Which numbers between 100 and 200 are perfect squares?
- Ans: 100, 121, 144, 169 and 196.

Q7. Is 150 a perfect square?
- Ans: No. sqrt(150) is about 12.25, which is not a whole number.

Q8. Between which two whole numbers does sqrt(180) lie?
- Ans: Between 13 and 14 (since 13x13 = 169 and 14x14 = 196).

Q9. Find the value of sqrt(169) + sqrt(144).
- Ans: 13 + 12 = 25.

Q10. A square park has an area of 196 square metres. Find the length of one side.
- Ans: Side = sqrt(196) = 14 metres.

Quick tip: To check if a number is a perfect square, find its square root.
If the root is a whole number, it is a perfect square; otherwise it is not.